CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 11: **Earth, Moon and the Sun** | Quick revision, one page

Rotation and revolution of the Earth

- Rotation: spinning of the Earth on its axis from west to east once in about 24 hours; causes day and night.
- Revolution: motion of the Earth around the Sun in about $365\frac{1}{4}$ days (one year).
- Leap year: 366 days; 29 February is added every 4 years to account for the extra quarter day.

Seasons

- The Earth's axis is tilted by about 23.5°; seasons are caused by this tilt and the revolution, not by the Earth's distance from the Sun.
- Around 21 June: longest day in the northern hemisphere; around 22 December: shortest day.
- Equinoxes (about 21 March and 23 September): day and night are nearly equal everywhere.
- When it is summer in the northern hemisphere, it is winter in the southern hemisphere.

The Moon and its phases

- The Moon is the natural satellite of the Earth; it shines by reflecting sunlight and takes about 27 days to revolve around the Earth.
- Phases: new moon (not visible), crescent, half moon, gibbous, full moon; the cycle repeats in about 29 to 30 days.
- Waxing: the bright part increases; waning: the bright part decreases.

Eclipses

- Solar eclipse: the Moon comes between the Sun and the Earth; occurs on a new moon day.
- Lunar eclipse: the Earth comes between the Sun and the Moon; occurs on a full moon day.
- Never look at the Sun directly or through a telescope during a solar eclipse; use proper filters.

MASTERMIND